ALAN TSENG

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EDUCATION

Master of Computer Science, National Taiwan University 2021-2024
Relevant Coursework: Blockchain and Big Data, Introduction to FinTech, Multimedia Security, Decentralized Applications Design and Practice, Computer Vision.
Overall GPA: 4.12/4.3

EXPERIENCE

Blockchain Developer Intern 2022 - 2022
Line Corporation (95M MAUs worldwide, social media company in Asia) Taipei, Taiwan
(Blockchain, React, Nx, Kubernetes, Swagger, Python)

- Swagger TypeGen Service
  - Situation: The existing API development process was inefficient and lacked a single source of truth.
  - Action: Design a service utilizing Swagger, auto-gen types, and Kubernetes deploy with DNS/LB.
  - Result: Increased API development speed by XX%.
- Built a scalable frontend using Nx in monorepo and collaborated with the Korean backend team.
- Hackathon: Led 5 developers to build an NFT merchant management system (NextJS/Spring Boot).
- Sharing: Gave internal talk on Ethereum & dApp development.

PROJECTS

Decentralized Ticketing System, Web3 Project
Master's Thesis (Solidity, TypeScript, React Native, Python, IPFS, GunDB, SQL, NextJS)

- Situation: Scalper raising prices leads to inflation, hurts event revenue, and prevents fans from buying tickets.
- Action: Simulate customer response to 3 ticketing strategies, analyzing stakeholder gaming for each.
- Result:
  - Reduced price fluctuations by 9X%, eliminating scalpers profit (1XX%)
  - Decentralized Ticketing App with 9X% BDD code coverage.

Rate My Professors TW, Full Stack Project
Side Project (NextJS, MVC, JWT, CI/CD, Web Crawling, NodeJS, MongoDB, PWA)

- Data: Increased the crawling speed by 8X% using asyncio and aiohttp.
- Backend: Initially developed with Express MVC, later migrated to FastAPI Swagger.
- Frontend: Initially used Pug, now using NextJS with course scheduler, rating system, and PWA support.

Drum.io, Game
Course Project (JavaScript, C, SocketIO, Arduino, Tensorflow.js)

- Developed a online multiplayer air drum game, using ML, Web Socket and Arduino.
- Result: Got 1st place in Interactive Media Programming (50+) by collective vote.

Create your dancing robot, Computer Graphics
Course Project (C/C++, OpenGL)

- Developed a Maya like model control system that allows users to record and play animations.
- Result: Got First place in the Computer graphics final project in a collective vote of 52 people.

Landing Planet, Game
Course Project (Java, Android)

- Won second place in a space exploration game development competition, with the support of 50 participants.
- Developed a space exploration game using Java and object-oriented concepts.

EXTRA-CURRICULAR ACTIVITIES

Developer Student Club Leader (2019-2021) - Advisor by Yuh-Pyng, Shieh, and Google Developers

- Chosen as top 800 leader from 3000+ global applicants for 2-year peer learning program.
- Led the club, engaging over 73 students with talks on Deep Learning, PoseNet, Python & Colab, and Flutter.

ACM Competitive Programing Club Leader (2017-2021) - Supervised by Jeng-Sheng Yeh

- Enhanced programming skills through practice and problem-solving on platforms like UVa Online Judge.

Volunteered at exhibitions

- Provided technical support at the ART FUTURE exhibition showcasing blockchain applications.
- Assisted with Developer Student Club (DSC) and Google Developer Groups (GDG) events at SITCON(Student Information Technology Conference) and COSCUP(Conference for Open Source Coders, Users and Promoters).

HONORS

Research Excellence Award

- Recognized for outstanding research contributions, placing among the top 1% of graduating students.

Intel Andy Grove Scholarship

- Recognized as a top scholar (18%, 367 selected) for my academic achievements and potential.

College Student Research Scholarship

- Awarded a prestigious scholarship by the Ministry of Science and Technology, Taiwan.

PUBLICATIONS

- Sung, H.-M., Chen, T., **Tseng, H.-C.**, Prayogo, B., Lin, J.-Y., & Hung, Y.-P. (2023). **akaTick: Hybrid Mobile E-Ticketing System Based on Non-Fungible Tokens**. 2023 IEEE International Conference on Metaverse Computing, Networking and Applications (MetaCom), 686–687.
- **Tseng, H.-C**., Tu, C.-W., Huan, X.-Y., & Chia, T.-L. (2020). **Nighttime vehicle light detection based on deep learning and image enhancement**. 2020 IPPR Conference on Computer Vision, Graphics, and Image Processing (CVGIP), 1–8.

SKILLS

- Blockchain(EVM Compatible Chains(L1/L2), Tezos, COSMOS, Solana), Artificial Intelligence(Machine Learning, DeepLearning), Web(Frontend/Backend) Development, Cloud-Native Applications, Computer Vision, Socket, Chrome Extension, SSG/SSR, MVC,
- Python, C/C++, JavaScript/TypeScript, Solidity, Java, SmartPy, Swift, Dart, C#, Matlab
- FastAPI/Flask, NodeJS, React(NextJS, React Native), Tensorflow, Scikit-Learn, OpenCV, OpenGL, Numpy, Pandas, Matplotlib, Processing(p5), Flutter
- NoSQL(MongoDB), SQL(PostgreSQL, MySQL), p2p(orbitDB, GunJS), IPFS